

Subject - Chemistry

* Section A1. The IUPAC name of $(CH_3)_2CH(CH_2)CH_2CH_3$ is:

- ✓ a. 2-methylbutane
 b. 3-methylbutane
 c. 2-ethylpropane
 d. pentane.

2. In the compound $(CH_3)_3C-OH$, the carbon attached to oxygen is:

- a. primary
 b. Secondary
 c. Tertiary
 d. Quaternary

3. Which pair represents chain isomerism?

- a. propan-1-ol and propan-2-ol
 ✓ b. Butane and 2-methylpropane
 c. Ethanal and dimethyl ether
 d. cis-2-butene and trans-2-butene

4. Which compounds can show geometrical isomerism?

- a. propene
 b. 1-butene

✓ c. 2-buten-2-ol

d. 2-methylpropene

5. The rate of S_N1 reaction depends on:
a. Substrate Concentration only.

6. Which substrate undergoes S_N1 reaction most readily?

✓ a. $(CH_3)_3C-Cl$

7. S_N2 reaction are favoured by

✓ b. strong, unhindered nucleophile

8. $E2$ elimination is

✓ b. bimolecular and one step

9. which mechanism generally involves a carbon intermediate?

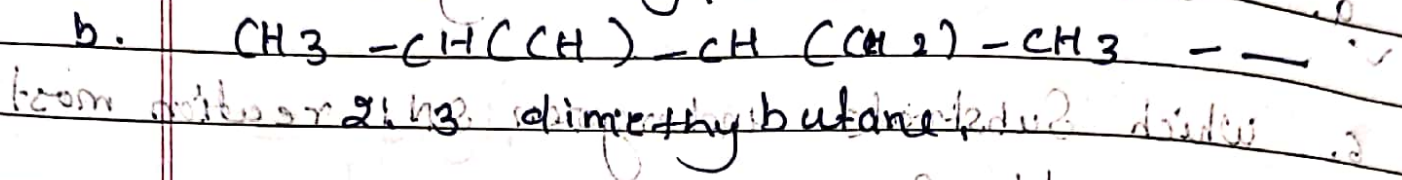
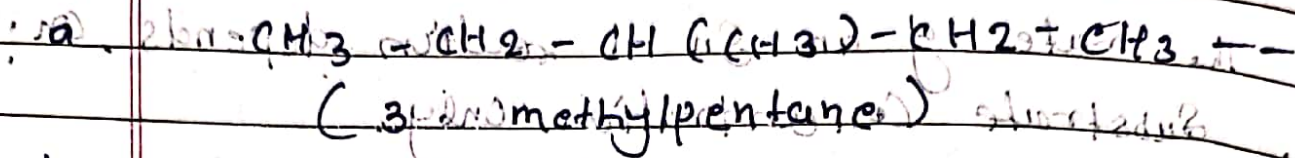
c. S_N1

10. According to Saytzeff's rule, the major alkene is generally

✓ b. The most substituted alkene.

* Section B - Short answer

11. Give the IUPAC names for:



12. Define the following with one example each:

a. primary Carbon

b. Secondary Carbon

c. Tertiary Carbon

d. Quaternary Carbon

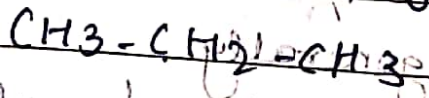
→ The degree of carbon atom is determined by the number of other carbon atoms directly attached to it.

a. primary Carbon (1°)

→ attached to one carbon atom

example: The middle CH_2 Carbon is Secondary (X)

The terminal CH_3 Carbons are primary ✓



→ Secondary Carbon (2°):

attached to two Carbon atoms
example The middle CH_2 Carbon is Secondary
 $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$

Tertiary Carbon (3°)

→ attached to three Carbon atoms
example The central C Carbon is tertiary
 $(\text{CH}_3)_3\text{CH}$

Quaternary Carbon (4°)

→ attached to four Carbon atoms
example The central Carbon is Quaternary
 $(\text{CH}_3)_4\text{C}$

13. What is structural isomerism? explain chain isomerism and position isomerism with example

→ Structural isomers are compounds having the same molecular formula but different

Connectivity of C atoms

(a) Chain isomerism

→ different Carbon skeletons

example $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$ butane

and $(\text{CH}_3)_3\text{CH}$ methylpropane

both have molecular formula C_4H_{10}

(b) Position isomerism. carb & functional group
 → Same carbon skeleton but different positions of the functional group or multiple bond
example: $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{OH}$ propan-1-ol
 $\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH}_3$ propan-2-ol

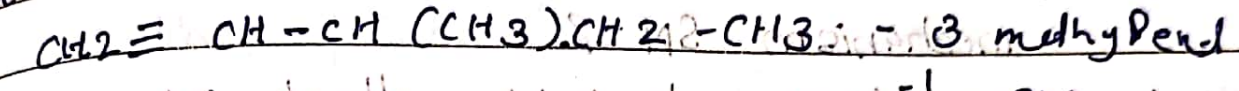
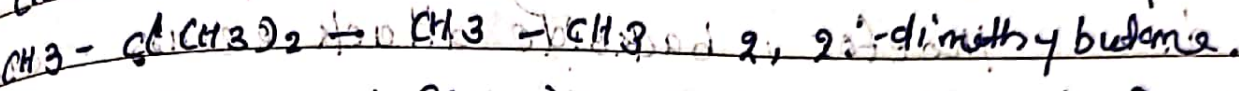
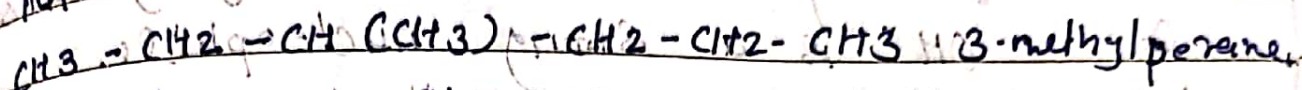
Q. 14. Differentiate between $\text{S}_\text{N}1$ and $\text{S}_\text{N}2$ reaction with respect to

- mechanism
- Rate law
- Substrate (primary, secondary, tertiary)
- Stereo chemistry

property	$\text{S}_\text{N}1$	$\text{S}_\text{N}2$
Meaning	Substitution nucleophilic unimolecular	Substitution nucleophilic bimolecular
Steps	Two steps	One step
Intermediate	Yes (Carbocation)	No
Rate law	$\text{Rate} = k[\text{R-X}]$	$\text{Rate} = k[\text{R-X}][\text{Nuc}]$
Favored substrate	$3^\circ > 2^\circ > 1^\circ$	$\text{CH}_3 > 1^\circ > 2^\circ$
Nucleophile	Weak nucleophile	Strong nucleophile preferred
Solvent	Polar protic	Polar aprotic preferred
Stereochemistry	Racemisation	Inversion of configuration

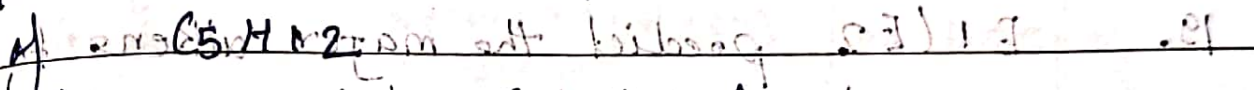
Section C - Application based: Question.

IUPAC Nomenclature: Give IUPAC names:

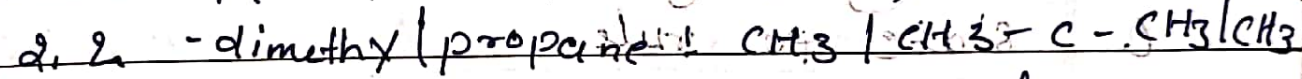
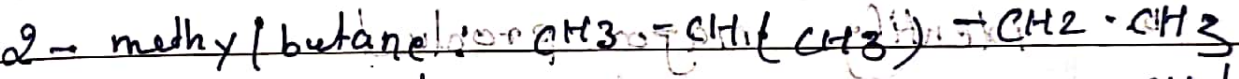
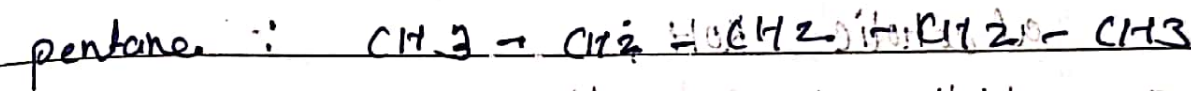


Isomerism.

Draw and name all possible constitutional isomers



There are three constitutional isomers.



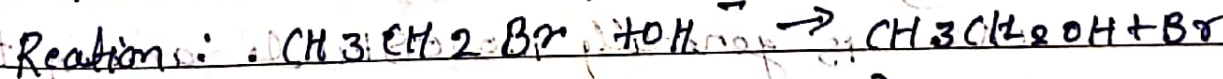
therefore: C_5H_{12} has 3 constitutional isomers.

Q.18 $\text{SN}_1 / \text{SN}_2$ predict the major product and

mechanism when $\text{CH}_3\text{CH}_2\text{Br}$ reacts with OH^-

Explain why the reaction follows SN_2 reaction

than SN_1 .

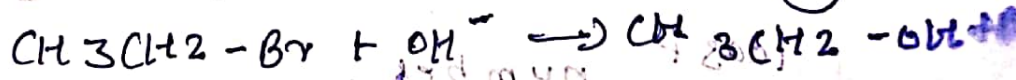


product: ethanol ($\text{CH}_3\text{CH}_2\text{OH}$)

Mechanism: OH^- attacks the carbon

bearing Br from the back side, while the

C-Br bond breaks simultaneously.



* why SN2?

q. Ethyl bromide is a primary alkyl halide.

b. primary carbocation are highly unstable.

c. OH⁻ is a strong nucleophile.

d. therefore, backside attack occurs readily.

mechanism SN-2

d. therefore, backside attack occurs readily.

mechanism SN-2

19. E1/E2 predict the major alkene formed when 2-bromobutane is treated with alcoholic KOH.

q. Write the reaction

identify major product

state the rule

Reaction

Reaction: $\text{CH}_3\text{CH}_2\text{Br} \rightarrow \text{CH}_2=\text{CH}_2$ alkene

the possible products are

major product: But-2-ene $\text{CH}_3\text{CH}=\text{CHCH}_3$

minor product: But-1-ene $\text{CH}_2=\text{CHCH}_2\text{CH}_3$

Among the but-2-ene isomers, trans-but-2-

ene is generally more stable than cis-but-2-ene.

Reason: The reaction follows Saytzeff's rule.

The major product is generally more stable.

alkene. Beta H forms a double bond with

at least 1 carbon.

Q.16 what is difference between $E1$ and $E2$ elimination reaction?

property	$E1$	$E2$
No. of steps	Two	one
Rate	$k[RX]$	$k[RX][Base]$
Intermediate	Carbocation	None
Base	weak base	strong base generally required
Substrate	$3^\circ > 2^\circ$	$3^\circ > 2^\circ > 1^\circ$ depending on base
Rearrangement	possible	not possible
Stereochemistry	not strictly required	only possible with periplanar geometry

Section D: Long answer / mechanism

Q.20 Explain the complete mechanism of an S_N1 reaction using tert-butyl bromide as an example.

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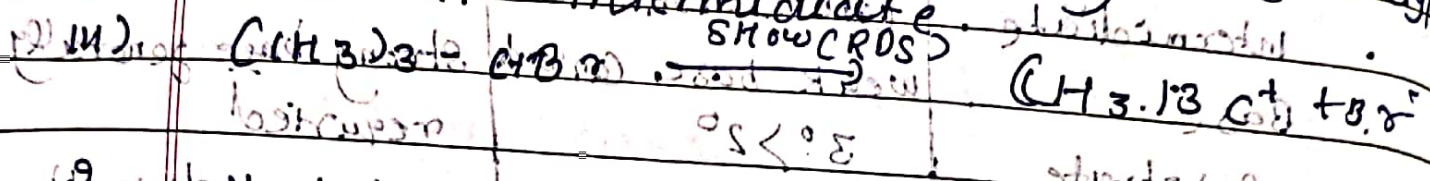
1. Ionisation / ionisation step
2. Carbocation formation
3. Nucleophilic attack
4. Rate determining step
5. Stereochemical outcome

Overall reaction: $(CH_3)_3C-Br + OH^- \rightarrow (CH_3)_3C-OH + Br^-$

substitution mechanism step by step mechanism of SN1

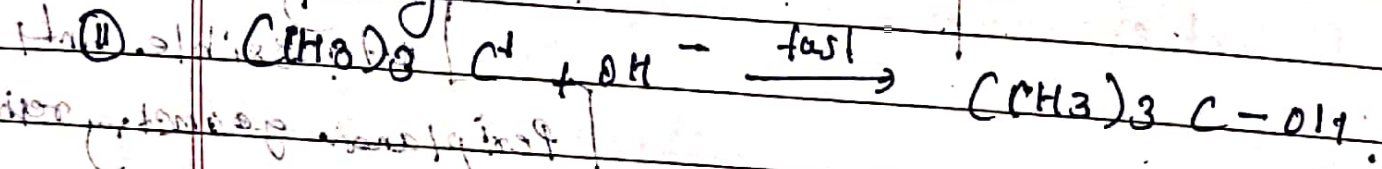
1. Ionization & Carbocation Formation (Rate Determining Step)

(i) The C-Br bond breaks slowly
(ii) This forms a stable tertiary carbocation intermediate



2. Nucleophilic Attack

(i) The nucleophilic (OH^-) attacks the carbocation rapidly



stereochemical Outcome

The intermediate carbocation is trigonal planar (sp^2)

The nucleophile can equally attack from the front or back leading to racemization (equal parts inversion and retention) if the starting molecule is chiral

Q.21

Compare SN_1 , SN_2 , E_1 and E_2 reactions in a suitable table



Discuss the important physical and chemical properties of alkyl halides.

Number of steps

Rate Law

Intermediate

Substrate preference

Nucleophile / base requirement

Solvent preference

Stereochemical requirement

Typical products

Feature	SN1	SN2	E1	E2
Step intermediate	2	1	2	1
Intermediate	Carbocation	None	Carbocation	None
Rate Law	$k[RX]$	$k[RX][Nu^-]$	$k[RX]$	$k[RX][Base]$
Substrate	$3^\circ > 2^\circ > 1^\circ$	$1^\circ > 2^\circ > 3^\circ$	$3^\circ > 2^\circ > 1^\circ$	$3^\circ > 2^\circ > 1^\circ$ depending on base
Nucleophile/base	Weak nucleophile possible	Strong nucleophile	Weak base	Strong base
Rearrangement	Possible	No	Possible	No
Product	Substitution	Substitution	Alkene	Alkene
Stereo effect	Less imp	Very imp	Less imp	Imp
Stereochemistry	Racemisation tendency	Inversion	Striked requirement	Strong base thermal
Typical Condition	Polar protic	Polar aprotic	Polar protic	Thermal

Q.22 Arrange the following in increasing order of S_N1 reactivity

CH_3Br , CH_3CH_2Br , $(CH_3)_2CHBr$, $(CH_3)_3CBr$
Give a reason for your answer.

Ans. 22. Increasing S_N1 Reactivity
Compounds - CH_3Br , CH_3CH_2Br , $(CH_3)_2CHBr$, $(CH_3)_3CBr$
increasing order
 $CH_3Br < CH_3CH_2Br < (CH_3)_2CHBr < (CH_3)_3CBr$
Give increasing order -
 $CH_3Br < CH_3CH_2Br < (CH_3)_2CHBr < (CH_3)_3CBr$

Methyl < primary < secondary < Tertiary
Reason:

S_N1 involves carbocation formation

Carbocation stability follows
 $3^\circ > 2^\circ > 1^\circ > \text{methyl}$ because alkyl groups stabilize the positive charge through +I effect and hyperconjugation.

charge through +I effect and hyperconjugation.
therefore, tertiary alkyl bromide undergoes S_N1 most readily.

Section F - Hybridisation - multiple choice

Questions.

Q.1 The hybridisation of Carbon in methane (CH_4) is,
Ans. (c) sp^3

Q.2 The hybridisation of Carbon atom in ethene ($\text{CH}_2=\text{CH}_2$) is

Ans. (b) sp^2

Q.3 The hybridisation of Carbon atom in ethyne ($\text{HC}\equiv\text{CH}$) is

Ans. (a) sp

Q.4 The shape of an sp^3 -hybridised carbon atom is

Ans. (c) Tetrahedral.

Q.5 The approximate bond angle in an sp^2 -hybridised Carbon is,

Ans. (c) 120°

Q.6 The approximate bond in an sp hybridised Carbon.

Ans. (d) 180°

Q.7 How many hybrid orbitals are formed in sp

Ans. (c) 4.

Q.8 How many unhybridised p orbitals are present in an sp^2 -hybridised Carbon atom?

(b) 1

Q.9 A π bond is formed by

(C) Sideways overlap of p orbitals

Q.10 which of the following contains an sp-hybridised Carbon atom?

(C) Ethyne